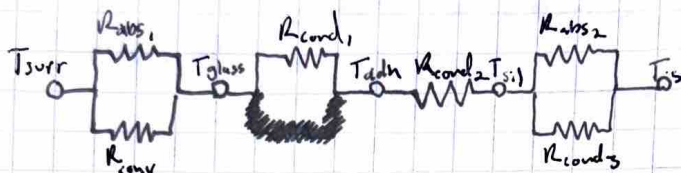


HW #4 Heat Transfer

Given:

2)



Assumptions: 1D heat flux

- No irradiation into adhesive, glass
- S.S.

(per area)

$$R_{rad1} = \frac{1}{h_{rad}} = 0.2$$

$$R_{conv} = \frac{1}{h} = 0.1$$

$$R_{cond1} = \frac{L_1}{k} = 0.002$$

$$R_{cond2} = \frac{L_2}{k} = 0.0001$$

$$R_{rad2} = \frac{1}{h_{rad}} = 0.2$$

$$R_{cond3} = \frac{L_3}{k} = 10^{-4}$$

b)

$$q''_{abs1} = 0.16 = \boxed{80 \frac{W}{m^2}}$$

$$q''_{enc} = \eta(680)$$

$$q''_{abs2} = \boxed{680 \frac{W}{m^2}}$$

$$T_{glass} = T_{surr} + q''_{out} R_{ext}$$

$$\boxed{341.1 K} \quad T_{sil} = T_{glass} + q''_{sil} R_{sil} = \boxed{342.3 K}$$

$$c) \quad q = a - bT_{sil} = 0.553 - T_{sil} = \boxed{0.2107}$$

$$q''_{enc} = \eta(680) = \boxed{143 \frac{W}{m^2}}$$

2)

$$2) \quad m = \sqrt{\frac{hP}{KA_c}}$$

In insulated part of rod, acts like 1D conduction.

$$q''_{ins} = \frac{KA_c}{L_{ins}} (T_w - T_o) = \frac{80(0.005^2 \pi)(473.15 - T_o)}{0.2m}$$

$$q''_{exposed} =$$

B.C.s:

$$① \quad -k \frac{dT}{dx} \Big|_{x=L} = 0$$

$$② \quad T(0) = T_o$$

$$T(x) - T_o = C_1 \exp\left(\sqrt{\frac{hP}{KA_c}} x\right) + C_2 \exp\left(-\sqrt{\frac{hP}{KA_c}} x\right)$$

$$T'(x) = \sqrt{\frac{hP}{KA_c}} \left[C_1 \exp\left(\sqrt{\frac{hP}{KA_c}} x\right) - C_2 \exp\left(-\sqrt{\frac{hP}{KA_c}} x\right) \right]$$

$$T_o = C_1 \exp\left(\sqrt{\frac{hP}{KA_c}} x\right) + C_2 \exp\left(-\sqrt{\frac{hP}{KA_c}} x\right)$$

Solving:

$$C_1 = \theta_b$$

$$C_2 = -\theta_b \tanh(mL_o)$$

$$q = \sqrt{hPKA_c} (T_o - T_w) \tanh(mL_o)$$

b) find T_o :

$$T_o = \frac{(0.03142)(200) + (0.0606)(20)}{0.03142 + 0.0606} = 81.5^\circ\text{C} = \boxed{354.65^\circ\text{K}}$$

$$L_o = 200\text{mm meets limit}$$

c) Min L_o ?

$$\frac{KA_c}{L_{ins}} (T_w - T_o) = C_o \tanh(mL_o) (T_o - T_o)$$

$T_o = 100^\circ\text{C}$; plug in:

$$\boxed{L_o = 0.073\text{m}}$$

3.

Given:

$$K = 400 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

convective tp

$$s = 0.01 \text{ m}$$

$$h = 100 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$h_f = 0.6$$

2)

BC's:

$$-k \frac{dT}{dx} \bigg|_{x=L} = h(T(L) - T_\infty)$$

$$A_{fin} = .01^2 = .0001$$

$$m = \sqrt{\frac{hP}{KA_c}} \quad \theta(0) = \theta_b$$

$$T(x) - T_\infty = c_1 \exp(mx) + c_2 \exp(-mx)$$

$$c_1 = \theta_b \quad c_2 = -\theta_b$$

$$\sqrt{hPKA_c} \theta_b \frac{\sinh mL + \frac{h}{mk} \cosh mL}{\cosh mL + \frac{h}{mk} \sinh mL} \quad \eta = \frac{h(PL + KA_c)}{h(PL + KA_c)}$$

Plug in .0.6 & numbers, solve:

$$L = 0.1487 \text{ m}$$

$$b) R_f = \frac{T_b - T_\infty}{q_f} = \frac{KA_c m \frac{h}{mk} \cosh mL + \sinh mL}{\cosh mL + \frac{h}{mk} \sinh mL} = 0.3629 \frac{\text{W}}{\text{K}}$$

$$c) \varepsilon = \frac{q_f}{hKA_c} = 36.3$$